

Proof of conjecture 00000003287: Gauss–Christoffel quadrature has its nodes at the zeros of the orthogonal polynomial and maximal precision $2n - 1$

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2026-10-05

1 The conjecture and its reading

Statement (English, verbatim). Definition: Orthogonal polynomials: three-term recurrence families of weight functions. Conjecture: Christoffel numerical integration: the nodes of Gauss quadrature are polynomial zeros, with the algebraic precision of quadrature the maximum $2n - 1$. (Gauss quadrature $2n-1$ maximal precision.) The Chinese text states the same.

This is the classical theorem on Gauss–Christoffel quadrature; we prove it (it is a known result, not new). Quotations below are transliterated to ASCII. The Wikipedia article “Gaussian quadrature” (retrieved in this session) states that an n -point Gaussian rule “is a quadrature rule constructed to yield an exact result for polynomials of degree $2n - 1$ or less by a suitable choice of the nodes x_i and weights w_i for $i = 1, \dots, n$ ”, and, in its proof of the fundamental theorem, “Therefore p_n has n distinct roots, all real, in the interval from a to b .”

Setting. We work in the most general setting we could make faithful: a measure μ on $\mathbb{R}$ such that

- (**FiniteMoments**) $x \mapsto x^k$ is μ -integrable for every $k \in \mathbb{N}$ (in particular μ is finite);
- (**NotFinitelySupported**) $\mu(\mathbb{R} \setminus s) \neq 0$ for every finite set s , i.e. μ has infinitely many points of increase.

The second condition is the standard non-degeneracy condition: the Wikipedia article “Orthogonal polynomials” says the inner product $\langle f, g \rangle = \int fg d\alpha$ “is positive definite if the function α has an infinite number of points of growth”. A weight function $w \geq 0$ on an interval $[a, b]$ (or on $\mathbb{R}$) is the case $\mu = w(x) dx$ with $w = 0$ off the interval; Lean lemma `notFinitelySupported_withDensity` proves that $w(x) dx$ satisfies the second condition whenever $\int w \neq 0$. The moment condition is kept as a hypothesis.

Definitions (all as in the Lean file).

- p is the *monic orthogonal polynomial of degree n* (**IsMonicOrthogonal**) if p is monic, $\deg p = n$, and $\int pq d\mu = 0$ for every real polynomial q with $\deg q < n$ (the zero polynomial has degree $-\infty$ and is included).
- An *n -node rule* is a pair of maps $x, w : \{0, \dots, n - 1\} \rightarrow \mathbb{R}$ (nodes and weights; arbitrary, not necessarily distinct nodes, arbitrary real weights) and the functional $f \mapsto \sum_i w_i f(x_i)$. It is *exact up to degree d* (**IsExactUpTo**) if $\int f d\mu = \sum_i w_i f(x_i)$ for every polynomial f with $\deg f \leq d$ (Lean’s `natDegree`; the zero polynomial is included).

- The *algebraic precision* (`algebraicPrecision`) is $\sup\{d \in \mathbb{N} : \text{the rule is exact up to degree } d\}$ (Lean’s `sSup` on $\mathbb{N}$; for the Gauss rule we prove this set has greatest element $2n - 1$, so no convention for empty or unbounded sets is used).
- The *Christoffel numbers* of nodes x (`christoffelWeight`) are $\lambda_i = \int \ell_i d\mu$, where $\ell_i = \prod_{j \neq i} (X - x_j)/(x_i - x_j)$ is the Lagrange basis polynomial (Mathlib’s `Lagrange.basis`).

Reading. With $n \geq 1$ (natural-number subtraction $2n - 1$ is then ordinary subtraction):

1. the orthogonal polynomial p_n exists and is unique, and the “three-term recurrence family” of the Definition is realized: $p_{n+2} = (X - a)p_{n+1} - bp_n$ with $b > 0$;
2. p_n has n real zeros, all simple;
3. the Gauss rule (nodes the zeros $x_1 < \dots < x_n$ of p_n , weights the Christoffel numbers) has positive weights, is exact up to degree $2n - 1$ and not exact up to degree $2n$, so its algebraic precision is exactly $2n - 1$;
4. “the nodes of Gauss quadrature are polynomial zeros”: every n -node rule that is exact up to degree $2n - 1$ satisfies $p_n = \prod_i (X - x_i)$, so its nodes are exactly the zeros of p_n ;
5. “the maximum $2n - 1$ ”: every n -node rule (any nodes, any weights) exact up to degree d has $d \leq 2n - 1$.

2 Proof

Write $I(f) = \int f d\mu$; by the moment condition every polynomial is integrable, so I is a linear functional on $\mathbb{R}[X]$ (Lean: `integrable_eval`, `I`).

Lemma 1 (`integral_pos`). *If $f \neq 0$ and $f(x) \geq 0$ for all $x \in \mathbb{R}$, then $I(f) > 0$.*

Proof. $I(f) \geq 0$. If $I(f) = 0$ then $f = 0$ μ -a.e., so μ is concentrated on the finite zero set of f , contradicting `NotFinitelySupported`. $\square$

In particular $I(r^2) = 0$ forces $r = 0$ (`eq_zero_of_I_mul_self`).

Existence and uniqueness (`exists_monicOrthogonal`, `monicOrthogonal_unique`). Let V be the polynomials of degree $< n$, identified with $\mathbb{R}^n$ by coefficients. The linear map $\Phi : V \rightarrow \mathbb{R}^n$, $\Phi(r)_i = I(rX^i)$, is injective: $\Phi(r) = 0$ gives $I(rq) = 0$ for all $q \in V$, hence $I(r^2) = 0$ and $r = 0$. So Φ is surjective, and choosing r with $\Phi(r)_i = -I(X^n X^i)$ gives the monic orthogonal $p_n = X^n + r$. If p, p' are both monic orthogonal of degree n , then $d = p - p'$ has degree $< n$ and $I(d^2) = I(pd) - I(p'd) = 0$, so $p = p'$.

Real simple zeros (`roots_card_eq`, `roots_nodup`). Key observation (`not_nonneg_mul`): there is no $s \neq 0$ with $\deg s < n$ and $p_n s \geq 0$ on $\mathbb{R}$, since then $0 < I(p_n s) = 0$. Write $p_n = Pq$ with $P = \prod_a (X - a)$ over the real roots (with multiplicity) and q without real zeros (Mathlib `exists_prod_multiset_X_sub_C_mul`). By the intermediate value theorem q has constant sign (`const_sign`). If P had degree $< n$, then $s = \pm P$ would give $p_n s = \pm qP^2 \geq 0$, impossible; so p_n has n real roots with multiplicity. If some root a had multiplicity ≥ 2 , then $p_n = (X - a)^2 s$ with $\deg s = n - 2$ and $p_n s = (X - a)^2 s^2 \geq 0$, impossible; so the roots are simple.

Exactness up to $2n - 1$ (`exact_of_roots`). Let $x_1, \dots, x_n$ be the zeros of p_n and $\deg f \leq 2n - 1$. Divide: $f = p_n d + r$ with $\deg r < n$ and $\deg d \leq n - 1$ (Mathlib `modByMonic_add`).

div). Then $I(p_n d) = 0$, $r = \sum_i r(x_i) \ell_i$ (Lagrange interpolation, `Lagrange.eq_interpolate`) and $f(x_i) = r(x_i)$, so $I(f) = I(r) = \sum_i r(x_i) \lambda_i = \sum_i \lambda_i f(x_i)$.

Positive weights (`weight_pos`). For any rule with distinct nodes that is exact up to degree $2n - 1$, apply exactness to ℓ_i^2 (degree $2n - 2$): $w_i = \sum_j w_j \ell_i(x_j)^2 = I(\ell_i^2) > 0$.

Maximality (`not_exact_two_mul`, `precision_le`). For any n -node rule let $\omega = \prod_i (X - x_i)$, of degree n . Then ω^2 has degree $2n$, $I(\omega^2) > 0$, and the rule gives $\sum_i w_i \omega(x_i)^2 = 0$. So no n -node rule is exact up to degree $2n$, hence none is exact up to any $d \geq 2n$; i.e. every exactness degree is $\leq 2n - 1$. For the Gauss rule the set of exactness degrees therefore has greatest element $2n - 1$ (`precision_eq`).

Nodes are the zeros (`nodes_eq`). If an n -node rule is exact up to $2n - 1$, then for $\deg q < n$ the product ωq has degree $\leq 2n - 1$, so $I(\omega q) = \sum_i w_i \omega(x_i) q(x_i) = 0$. Thus ω is monic orthogonal of degree n , and by uniqueness $\omega = p_n$.

Three-term recurrence (`three_term_recurrence`). Let p_n, p_{n+1}, p_{n+2} be monic orthogonal. $Xp_{n+1} - p_{n+2}$ has degree $\leq n + 1$; subtracting $a p_{n+1}$ and then $b p_n$ (with a, b the appropriate top coefficients, `degree_sub_lead`) leaves v of degree $< n$ with $I(v X^i) = 0$ for $i < n$ (using $I(p_{n+1} X^{i+1}) = 0$), so $v = 0$: $p_{n+2} = (X - a)p_{n+1} - b p_n$. Finally $b I(p_n^2) = I(p_{n+1} \cdot X p_n) = I(p_{n+1}^2) > 0$, so $b > 0$.

3 Lean formalization and axiom audit

File `lean/Conjecture3287/Basic.lean` (namespace `C3287`; only import `Mathlib`). The main theorem is

```
theorem gauss_christoffel (hmom : FiniteMoments mu)
  (hsupp : NotFinitelySupported mu) {n : Nat} (hn : 0 < n) :
  (exists! p : R[X], IsMonicOrthogonal mu n p) /\
  forall p : R[X], IsMonicOrthogonal mu n p ->
    (Multiset.card p.roots = n /\ p.roots.Nodup) /\
    (exists x : Fin n -> R, StrictMono x /\
      (forall a : R, p.IsRoot a <-> a in Set.range x) /\
      (forall i, 0 < christoffelWeight mu x i) /\
      IsExactUpTo mu x (christoffelWeight mu x) (2 * n - 1) /\
      not IsExactUpTo mu x (christoffelWeight mu x) (2 * n) /\
      algebraicPrecision mu x (christoffelWeight mu x) = 2 * n - 1) /\
    (forall x w : Fin n -> R, IsExactUpTo mu x w (2 * n - 1) ->
      p = prod i, (X - C (x i))) /\
    (forall x w : Fin n -> R, forall d, IsExactUpTo mu x w d -> d <= 2 * n - 1)
```

(ASCII transliteration of the Lean statement). Further top-level results: `three_term_recurrence` (item 1) and `notFinitelySupported_withDensity` (weight functions $w(x) dx$ are admissible).

Axiom audit. `lake build` succeeds; `#print axioms for C3287.gauss_christoffel, C3287.three_term_recurrence and C3287.notFinitelySupported_withDensity` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no added axioms.

4 Sources

- Wikipedia, “Gaussian quadrature”, https://en.wikipedia.org/wiki/Gaussian_quadrature (quoted above).

- Wikipedia, “Orthogonal polynomials”, https://en.wikipedia.org/wiki/Orthogonal_polynomials (quoted above).